HAVING FINITELY MANY IRREDUCIBLE COMPONENTS IN EVERY QUASI-COMPACT OPEN IS LOCAL IN THE FPPF TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The property quantified over quasi-compact opens is the one used elsewhere in the chapter; the statement is reproduced verbatim including that quantifier.

Editorial note. The counted spans carry 6 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The locality criterion for a property of schemes in a topology. Same chapter, Lemma lemma-descending-properties, invoked at proof lines 5077 and 5095. Reduces locality in the fppf topology to Zariski locality together with the two statements about flat finitely presented morphisms of affines.
- Generic points lift along flat morphisms. Morphisms, Lemma morphisms-lemma-generalizations-lift-flat, invoked at proof lines 5081-5083. Allows the irreducible components of the source to be traced to those of the target plus those of the fibres.
- Fibres of a finitely presented morphism are of finite type over the residue field, and such schemes are Noetherian with finitely many irreducible components. Morphisms, Lemma morphisms-lemma-base-change-finite-type at proof lines 5085-5087, Morphisms, Lemma morphisms-lemma-finite-type-noetherian and Properties, Lemma properties-lemma-Noetherian-irreducible-components at proof lines 5088-5090. Bounds the number of generic points of each fibre.
- A surjective continuous map and irreducible components. Topology, Lemma topology-lemma-surjective-continuous-irreducible-components, invoked at proof lines 5093-5094. Descends the finiteness of the number of irreducible components from the source to the target.

Source reference. Original lemma 0349: sources/stacks-descent/pinned-original-source.tex, source lines 4918–4918. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 01T4: sources/stacks-chow/tags.txt, source lines 2361–2361. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T6: sources/stacks-chow/tags.txt, source lines 2363–2363. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03HV: sources/stacks-chow/tags.txt, source lines 4581–4581. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BA8: sources/stacks-chow/tags.txt, source lines 14201–14201. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GM2: sources/stacks-chow/tags.txt, source lines 20742–20742. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(S) =$"every quasi-compact open of S has a finite number of irreducible components" is local in the fppf topology.*

Proof. We will use Lemma 0349. First we note that $\mathcal{P}$ is local in the Zariski topology. Next, we show that if $T \to S$ is a flat, finitely presented morphism of affines and S has a finite number of irreducible components, then so does T. Namely, since $T \to S$ is flat, the generic points of T map to the generic points of S, see Morphisms, Lemma 03HV. Hence it suffices to show that for $s \in S$ the fibre T_s has a finite number of generic points. Note that T_s is an affine scheme of finite type over $\kappa(s)$, see Morphisms, Lemma 01T4. Hence T_s is Noetherian and has a finite number of irreducible components (Morphisms, Lemma 01T6 and Properties, Lemma 0BA8). Finally, we have to show that if $T \to S$ is a surjective flat, finitely presented morphism of affines and T has a finite number of irreducible components, then so does S. This follows from Topology, Lemma 0GM2. Thus (1), (2) and (3) of Lemma 0349 hold and we win. □